

Physics-Informed NoProp: Integrating Discovered Physical Laws into Localized Diffusion-Based Learning

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Abstract

Neural network training via back-propagation requires global gradient propagation across all layers, which imposes memory overhead, limits parallelism, and hinders biological plausibility. The recently proposed NoProp (No Propagation) framework circumvents these issues by decomposing the network into independently trained blocks, each learning to denoise a noisy target through a local diffusion process. While computationally attractive, NoProp lacks an explicit mechanism to incorporate prior physical knowledge about the system generating the data. In this paper, we propose a physics-informed NoProp framework that integrates a governing equation discovered by SPIDER (Sparse Physics-Informed Discovery of Empirical Relations) into blockwise learning. The framework combines the diffusion-based local learning paradigm of NoProp with weak-form sparse regression: the validated equation coefficients enter the model through a learned first-frame physical condition and local temporal-field constraints. We evaluate the framework using 15 independent 64^3 trajectories of unforced decaying homogeneous isotropic turbulence, with trajectory-disjoint discovery, validation, and test splits. SPIDER recovers the complete momentum equation with a maximum coefficient error of 0.368% and an independent test residual of 9.70×10^{-4} . In three-seed experiments, physics-informed NoProp achieves 86.73% and 82.94% mean accuracy in the low- and high-entrophy groups, compared with 17.59% and 17.46% for the architecture-matched no-equation NoProp control, while reducing the corresponding full Navier–Stokes residuals by 24.8% and 63.5%. The code is available at

<https://github.com/Flyhwjf/PI-NoProp>

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1. Introduction

Deep neural networks have achieved remarkable successes across a wide range of scientific and industrial domains, largely driven by the effectiveness of back-propagation (BP) for training multi-layer architectures [1]. BP iteratively adjusts network parameters by propagating an error signal from the output back through all layers, enabling end-to-end optimization of complex models. Despite its empirical success, BP suffers from several fundamental limitations. First, it is biologically implausible, as it requires symmetric forward and backward passes that do not align with local synaptic plasticity observed in the brain [2, 3]. Second, BP necessitates storing intermediate activations during the forward pass, imposing significant memory overhead that hampers scalability [1]. Third, the sequential nature of gradient propagation creates strong dependencies across layers, limiting parallelization and impeding efficient distributed training [4]. These drawbacks have motivated a sustained interest in developing alternative learning paradigms that do not rely on global error propagation [5, 6, 7, 8, 9].

Among these alternatives, **NoProp** (No Propagation) [10] has recently emerged as a promising framework that completely avoids forward and backward propagation across network blocks. Inspired by diffusion models [11, 12, 13], NoProp decomposes a neural network into independently trained blocks. During training, each block receives the input x and a noisy version of the target label embedding, and learns to denoise it using only local back-propagation within the block. At inference, Gaussian noise is progressively transformed through the blocks into a clean representation that is fed to a final classifier. By eliminating global gradient flow, NoProp enables blockwise parallel training, reduces memory footprint, and offers greater biological plausibility. Empirical results on MNIST, CIFAR-10, and CIFAR-100 demonstrate that NoProp achieves performance competitive with standard back-propagation on the same architectures [10].

However, NoProp, like most purely data-driven deep learning methods, does not explicitly incorporate any prior knowledge about the physical processes that generated the data. In many scientific applications – such as

fluid dynamics, climate modeling, or materials science – the observed data are governed by fundamental physical laws expressed as partial differential equations (PDEs). Ignoring this structure can lead to models that, while accurate on training data, produce predictions that violate physical constraints, especially under distribution shift or in the presence of noise [14]. This limitation is particularly acute when data are sparse or corrupted, a common scenario in experimental measurements.

Concurrently, a separate line of research has focused on **data-driven discovery of governing equations** from observational data. Starting with sparse regression methods like SINDy [15] for ordinary differential equations, numerous approaches have been developed to identify PDEs from spatiotemporal data [16, 17, 18, 19, 20, 21, 22]. A particularly robust framework is **SPIDER** (Sparse Physics-Informed Discovery of Empirical Relations) [23], which combines three key ingredients: (i) physical constraints of smoothness, locality, and symmetry to construct compact libraries of candidate terms; (ii) a weak formulation of PDEs that shifts derivatives onto test functions, drastically reducing sensitivity to noise; and (iii) sparse regression based on singular value decomposition to extract parsimonious relations. SPIDER has been successfully applied to highly turbulent channel flow data, recovering not only the Navier–Stokes and continuity equations but also the pressure–Poisson equation, energy equation, and boundary conditions, even under extreme noise levels [23].

Despite the power of SPIDER for equation discovery, it remains a post-hoc analysis tool that does not directly inform the training of neural networks used for downstream tasks such as prediction or classification. Conversely, NoProp offers an efficient learning engine but lacks a mechanism to enforce physical consistency. Bridging these two paradigms could yield models that are both computationally efficient and physically faithful, a goal that aligns with the emerging field of **physics-informed machine learning** [14, 24, 25, 26].

In this paper, we propose **physics-informed NoProp**, a framework that integrates a governing equation discovered by SPIDER into the blockwise training of NoProp. Our approach proceeds in two stages. First, we apply SPIDER to trajectory-disjoint observational data to recover and validate the complete incompressible momentum equation in four-dimensional weak form, yielding explicit and interpretable coefficients. Second, we transfer these coefficients into NoProp through two complementary paths. In the conditioning path, the coefficients combine first-frame convective, pressure, and viscous

energy-rate terms into a target-free physical condition that is encoded and fused into every NoProp block. In the local-constraint path, a pretrained and frozen temporal decoder reconstructs velocity and pressure time windows from each block latent z_t , and the weak momentum and continuity residuals augment that block’s local denoising objective without propagating gradients between blocks. The pressure–Poisson and kinetic-energy relations are algebraically derived from the discovered momentum equation and are evaluated only in the corresponding ablation variants.

We validate the proposed method on an independently generated dataset of unforced decaying homogeneous isotropic turbulence, comprising 15 trajectory-disjoint direct numerical simulations on 64^3 periodic grids. Our experiments demonstrate that physics-informed NoProp substantially improves future energy-decay classification over the architecture-matched no-equation NoProp control while producing latent reconstructions with lower Navier–Stokes residuals. Under controlled observation noise, the model remains predictive at a noise level of 0.1 channel standard deviations, although accuracy decreases in both enstrophy groups and approaches chance at higher noise levels. Ablation studies further evaluate the effects of the discovered momentum equation and the algebraically derived pressure–Poisson and kinetic-energy relations on predictive accuracy and physical consistency.

The main contributions of this work are threefold:

- We introduce a principled framework that combines the efficiency of NoProp’s blockwise diffusion-based learning with the interpretability of SPIDER’s data-driven equation discovery.
- We design a two-path physics interface that uses the discovered equation coefficients both in a learned first-frame physical condition and in a strictly local weak-form loss evaluated on frozen temporal-decoder reconstructions, without propagating gradients between NoProp blocks.
- We provide extensive empirical evidence on a challenging turbulent flow dataset, showing that the proposed method yields physically consistent representations, characterising its behaviour under controlled noise, and achieving strong predictive performance.

The remainder of this paper is organized as follows. Section 2 reviews the necessary background on NoProp and SPIDER. Section 3 details the proposed physics-informed NoProp framework. Section 4 presents experimental

setup, results, and discussion. Section 5 concludes the paper and outlines future research directions.

2. Related Work

This section reviews the literature most germane to the proposed physics-informed NoProp framework. We first discuss alternative learning paradigms that avoid global back-propagation, with a focus on the NoProp algorithm. Next, we survey data-driven methods for discovering partial differential equations (PDEs) from observations, emphasizing the SPIDER methodology. We then briefly cover physics-informed machine learning and diffusion-based generative models, which provide the theoretical underpinnings for our approach. Finally, we position our contribution in relation to these lines of work.

2.1. Local Learning and Alternatives to Back-Propagation

The canonical training algorithm for deep neural networks, back-propagation (BP) [1], relies on a global error signal that is propagated backwards through all layers. While highly effective, BP has several well-documented shortcomings: it is biologically implausible because it requires symmetric forward and backward passes [2, 3], it incurs a large memory footprint due to the need to store intermediate activations [1], and its sequential nature hinders parallelization [4].

To overcome these limitations, a variety of **local learning** schemes have been proposed. Early examples include deep belief networks trained layer-wise with contrastive divergence [5]. More recent methods include feedback alignment [3] and direct feedback alignment [7], which replace the symmetric weights of BP with fixed random matrices, yet still require a global error signal. Synthetic gradients [8] train auxiliary networks to approximate the gradient at each layer, allowing some decoupling. The forward-forward algorithm [9] trains each layer by contrasting the responses to positive and negative examples using a local objective.

Among these, **NoProp** (No Propagation) [10] stands out as a method that completely eliminates both forward and backward propagation across blocks. NoProp treats a neural network as a sequence of T blocks, each performing a denoising step inspired by diffusion models [11, 12, 13]. Formally, let x be an input and y its label, with associated embedding $u_y \in \mathbb{R}^d$ (either fixed one-hot or learnable). NoProp defines a forward (inference) process:

$$z_0 \sim \mathcal{N}(0, I), \quad z_t = a_t \hat{u}_{\theta_t}(z_{t-1}, x) + b_t z_{t-1} + \sqrt{c_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I), \quad (1)$$

for $t = 1, \dots, T$, where $\hat{u}_{\theta_t}$ is a neural network block and a_t, b_t, c_t are scalars derived from a noise schedule. The training objective for block t is derived from a variational lower bound on the conditional log-likelihood $\log p(y|x)$:

$$\mathcal{L}_{\text{NoProp}}^{(t)} = \frac{T}{2} \eta \mathbb{E}[(\text{SNR}(t) - \text{SNR}(t-1)) \|\hat{u}_{\theta_t}(z_{t-1}, x) - u_y\|^2], \quad (2)$$

where $\text{SNR}(t) = \bar{\alpha}_t / (1 - \bar{\alpha}_t)$ is the signal-to-noise ratio, η a hyperparameter, and the expectation is taken over the variational posterior. A final classifier $\hat{p}_{\theta_{\text{out}}}(y|z_T)$ is trained with cross-entropy. Crucially, each block can be updated independently using only local back-propagation, enabling massive parallelism and reducing memory consumption.

Despite its computational advantages, NoProp does not incorporate any prior knowledge about the physical laws that may govern the data. In scientific applications, such knowledge can be critical for generalization and robustness.

2.2. Data-Driven Discovery of Governing Equations

A parallel line of research aims to extract symbolic equations – particularly PDEs – directly from observational data. The **sparse identification of nonlinear dynamics** (SINDy) framework [15] pioneered this approach for ordinary differential equations by posing a sparse linear regression problem over a library of candidate functions. For PDEs, several extensions have been developed [16, 17, 18, 19, 20, 21, 22], but they often suffer from high sensitivity to noise because they operate on the **strong form** of the equations, which requires numerical differentiation of noisy data.

To mitigate noise sensitivity, **weak-form** formulations have been introduced [19, 20, 21]. The key idea is to multiply the PDE by a smooth test function w and integrate over a spatio-temporal domain Ω , shifting derivatives onto w via integration by parts. For a candidate relation of the form $\sum_n c_n f_n = 0$, this yields a linear system:

$$\sum_n c_n \int_{\Omega} w f_n d\Omega = 0, \quad (3)$$

where the integrals can be approximated numerically without ever differentiating the data. This weak formulation dramatically improves robustness, enabling accurate identification even with noise levels exceeding 100% [23].

The **SPIDER** (Sparse Physics-Informed Discovery of Empirical Relations) framework [23] synthesises these ideas into a comprehensive methodology. SPIDER incorporates three key ingredients:

1. **Symmetry-constrained library construction.** Libraries of candidate terms are built by considering irreducible representations of the symmetry group of the system (e.g., $O(3)$ for three-dimensional isotropic fluids). This yields compact libraries that respect rotational and reflectional symmetries, reducing the search space and improving interpretability. For example, for a fluid described by velocity $\mathbf{u}$ and pressure p , the scalar library up to third order is:

$$\mathcal{L}_0^{(3)} = \{1, p, \nabla \cdot \mathbf{u}, \partial_t p, p^2, u^2, p^3, \mathbf{u} \cdot \nabla p, \nabla^2 p, p \partial_t p, \partial_t^2 p, p \nabla \cdot \mathbf{u}, u^2 p, \mathbf{u} \cdot \partial_t \mathbf{u}\}, \quad (4)$$

and the vector library is:

$$\mathcal{L}_1^{(3)} = \{\mathbf{u}, \partial_t \mathbf{u}, \nabla p, p \mathbf{u}, (\mathbf{u} \cdot \nabla) \mathbf{u}, \nabla^2 \mathbf{u}, \partial_t^2 \mathbf{u}, u^2 \mathbf{u}, p^2 \mathbf{u}, \partial_t \nabla p, p \nabla p, \mathbf{u}(\nabla \cdot \mathbf{u}), (\nabla \mathbf{u}) \cdot \mathbf{u}, \nabla(\nabla \cdot \mathbf{u}), p \partial_t \mathbf{u}, u \partial_t p\}. \quad (5)$$

2. **Weak formulation and feature matrix construction.** For each term f_n in the library, its weak-form integral over many domains Ω_i (with different test functions w_i) is computed, forming a feature matrix $\mathbf{Q}$ with entries

$$Q_{in} = \frac{1}{V_i S_n} \int_{\Omega_i} w_i(\mathbf{x}, t) f_n(\mathbf{x}, t) d\Omega, \quad (6)$$

where V_i is a normalisation factor and S_n is a scale that non-dimensionalises the term.

3. **Sparse regression via singular value decomposition.** Relations of the form $\mathbf{c} \cdot \mathbf{f} = 0$ correspond to vectors $\mathbf{c}$ in the nullspace of $\mathbf{Q}$. SPIDER identifies parsimonious relations by examining the right singular vectors associated with the smallest singular values of $\mathbf{Q}$. A greedy algorithm then prunes terms to produce a sequence of increasingly sparse candidate equations, from which the most appropriate is selected based on a residual threshold.

In the cited SPIDER study, the method was applied to the Johns Hopkins turbulence database and recovered incompressible-flow relations under noisy observations [27, 23]. The present work instead reruns the discovery protocol on the independently generated HIT DNS described in Section 4.1.

2.3. Physics-Informed Machine Learning

The integration of physical knowledge into neural network training has gained substantial traction under the banner of **physics-informed neural networks (PINNs)** [14, 24]. PINNs embed the residual of a known PDE into the loss function, enabling the network to approximate solutions while respecting the governing physics. For a PDE $\mathcal{R}[\mathbf{u}] = 0$, the loss is typically

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{data}} + \lambda \|\mathcal{R}[\hat{\mathbf{u}}]\|^2, \quad (7)$$

where $\hat{\mathbf{u}}$ is the network output. This paradigm has been extended to inverse problems, parameter estimation, and discovery of unknown terms [25, 26]. However, PINNs assume the form of the PDE is known *a priori* – an assumption that limits their applicability in purely data-driven discovery scenarios.

2.4. Diffusion Models and Flow Matching

NoProp is deeply rooted in the theory of **denoising diffusion probabilistic models (DDPMs)** [11, 12, 13]. In DDPMs, a forward process gradually adds noise to data, and a neural network learns to reverse it. The training objective is a weighted sum of denoising score matching losses. The continuous-time limit leads to score-based generative models described by stochastic differential equations (SDEs) and probability flow ODEs [13]. **Flow matching** [38, 39] provides an alternative simulation-free training objective for continuous normalising flows, directly regressing a vector field that transports a prior to the data distribution. Both diffusion and flow matching have been adapted for supervised learning tasks such as classification and regression [35, 36, 37], demonstrating their versatility.

NoProp adapts these ideas to a discriminative setting by treating the label embedding as the target distribution. The forward (noising) process is fixed, and each block learns a conditional denoising step. The resulting framework is highly modular and parallelisable.

2.5. Positioning of This Work

The preceding discussion highlights a clear gap: NoProp provides an efficient local learning mechanism but lacks physical awareness, while SPIDER excels at discovering interpretable PDEs from data but is a post-hoc tool. The emerging field of physics-informed machine learning has shown the benefits of embedding physical constraints into neural networks, yet it typically assumes the governing equations are known. **Physics-informed NoProp**

bridges these worlds: it uses SPIDER to discover and validate the complete momentum equation directly from the available data, then incorporates its coefficients both as a predictive first-frame condition and as a weak-form local regulariser in the NoProp training objective. This synergy yields a model that preserves blockwise locality while improving predictive and physical consistency. To the best of our knowledge, this is the first attempt to combine blockwise diffusion-based learning with automated equation discovery in a unified framework.

3. Methodology

We present the proposed physics-informed NoProp framework, which transfers a SPIDER-discovered and independently validated momentum equation into the blockwise training of NoProp. Section 3.1 briefly recalls the NoProp algorithm [10] and establishes notation. Section 3.2 summarises the SPIDER methodology [23] for extracting interpretable physical relations from data. Section 3.3 describes the core contribution: the discovered coefficients enter each block through both a learned first-frame physical condition and a strictly local weak-form temporal-field loss. Section 3.4 gives the implementation details.

3.1. NoProp: Blockwise Diffusion-Based Learning

Let $x \in \mathcal{X}$ be an input (e.g., an image) and $y \in \{1, \dots, m\}$ its associated label. An embedding $u_y \in \mathbb{R}^d$ of the label is defined, either as a fixed one-hot vector or as a row of a learnable embedding matrix $W_{\text{Embed}} \in \mathbb{R}^{m \times d}$. NoProp [10] decomposes a neural network into T sequential blocks. During **inference**, a latent variable z_0 is drawn from a standard Gaussian prior $p(z_0) = \mathcal{N}(z_0; 0, I)$. For each block $t = 1, \dots, T$, the latent is updated by a residual diffusion step:

$$z_t = a_t \hat{u}_{\theta_t}(z_{t-1}, x) + b_t z_{t-1} + \sqrt{c_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I), \quad (8)$$

where $\hat{u}_{\theta_t} : \mathbb{R}^d \times \mathcal{X} \rightarrow \mathbb{R}^d$ is a neural network parameterised by θ_t , and a_t, b_t, c_t are scalar coefficients determined by a fixed noise schedule (e.g., cosine schedule). The final latent z_T is fed into a classifier $\hat{p}_{\theta_{\text{out}}}(y|z_T)$, typically a linear layer followed by softmax.

Training objective. NoProp is trained by maximising a variational lower bound on the conditional log-likelihood $\log p(y|x)$. A forward noising

process $q(z_t|y)$ is defined as an Ornstein-Uhlenbeck process that gradually adds noise to the label embedding u_y :

$$q(z_t|y) = \mathcal{N}(z_t; \sqrt{\bar{\alpha}_t} u_y, 1 - \bar{\alpha}_t), \quad (9)$$

where $\bar{\alpha}_t = \prod_{s=t}^T \alpha_s$ and α_s are related to the noise schedule. Using the reparameterisation trick, one can sample $z_{t-1} \sim q(z_{t-1}|y)$ directly. The evidence lower bound (ELBO) simplifies to a sum of per-block denoising score matching losses [10, 13]. For block t , the loss is:

$$\mathcal{L}_{\text{diff}}^{(t)} = \frac{T}{2} \eta \mathbb{E}_{q(z_{t-1}|y)} \left[(\text{SNR}(t) - \text{SNR}(t-1)) \|\hat{u}_{\theta_t}(z_{t-1}, x) - u_y\|^2 \right], \quad (10)$$

where $\text{SNR}(t) = \bar{\alpha}_t / (1 - \bar{\alpha}_t)$ is the signal-to-noise ratio and η is a hyperparameter. The classifier is trained with cross-entropy:

$$\mathcal{L}_{\text{cls}} = -\mathbb{E}_{q(z_T|y)} [\log \hat{p}_{\theta_{\text{out}}}(y|z_T)]. \quad (11)$$

The total NoProp loss (per sample) is $\mathcal{L}_{\text{cls}} + \sum_{t=1}^T \mathcal{L}_{\text{diff}}^{(t)}$ plus an often-neglected KL term $D_{\text{KL}}(q(z_0|y)||p(z_0))$ [10].

3.2. SPIDER: Discovering PDEs from Data

SPIDER [23] is a data-driven method for discovering interpretable PDEs from possibly noisy observations of spatiotemporal fields. Consider a physical system described by fields $\mathbf{u}(\mathbf{x}, t)$ (e.g., velocity) and $p(\mathbf{x}, t)$ (pressure). The method assumes that the governing equations can be written as linear combinations of candidate terms f_n – products of the fields and their partial derivatives – respecting the symmetries of the system. The goal is to find a sparse coefficient vector $\mathbf{c} = [c_1, \dots, c_N]^\top$ such that

$$\sum_{n=1}^N c_n f_n(\mathbf{u}, p, \partial_t \mathbf{u}, \nabla \mathbf{u}, \nabla p, \dots) = 0. \quad (12)$$

To avoid numerical differentiation of noisy data, SPIDER employs a **weak formulation**. Let Ω_i be a spatiotemporal domain and $w_i(\mathbf{x}, t)$ a smooth test function with compact support. Integrating (12) against w_i and integrating by parts where possible yields:

$$\sum_{n=1}^N c_n \int_{\Omega_i} w_i(\mathbf{x}, t) \tilde{f}_n(\mathbf{u}, p, \dots) d\Omega = 0, \quad (13)$$

where $\tilde{f}_n$ denotes the term after transferring derivatives onto w_i (if integration by parts is applicable) or the original term otherwise. Repeating this for many domains $i = 1, \dots, M$ and different test functions gives a linear system $\mathbf{Q}\mathbf{c} = \mathbf{0}$, with $\mathbf{Q} \in \mathbb{R}^{M \times N}$ having entries

$$Q_{in} = \frac{1}{V_i S_n} \int_{\Omega_i} w_i(\mathbf{x}, t) \tilde{f}_n(\mathbf{u}, p, \dots) d\Omega, \quad (14)$$

where $V_i = \int_{\Omega_i} |w_i| d\Omega$ is a normalisation factor and S_n is a scale that non-dimensionalises term n (estimated from characteristic scales of the data). The system is solved for $\mathbf{c}$ with $\|\mathbf{c}\| = 1$ by taking the right singular vector corresponding to the smallest singular value of $\mathbf{Q}$.

Sparse, parsimonious relations are obtained via a greedy algorithm that iteratively removes terms and tracks the residual $\|\mathbf{Q}^{(N)}\mathbf{c}^{(N)}\|$. The final selection balances accuracy and sparsity using a threshold on the relative increase in residual. The original SPIDER publication demonstrated this procedure on the Johns Hopkins database; here, the accepted artifact is independently discovered and validated on the generated HIT trajectories (Section 4.2.2).

3.3. Physics-Informed NoProp

Physics-informed NoProp couples equation discovery to local learning through a validated, machine-readable interface. The pipeline has three stages: (i) generate independent, densely sampled turbulence trajectories and identify a sparse four-dimensional relation; (ii) validate its support and coefficients on unseen trajectories and persist the complete provenance; and (iii) use the accepted relation both as a first-frame physical condition and in a weak-form temporal-field loss inside each isolated NoProp objective. An artifact that fails the discovery, support, coefficient, or held-out-residual checks is not admitted to training.

3.3.1. Discovering Governing Equations with SPIDER

For the unforced incompressible flow considered here, the target momentum relation is

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \nabla^2 \mathbf{u} = 0. \quad (15)$$

The vector candidate library contains the four terms in Eq. (15) and the same-order distractor $\nabla |\mathbf{u}|^2$. For a compactly supported time-space test func-

tion w , integration by parts gives the componentwise weak columns

$$\begin{aligned} q_i^{(t)} &= - \int u_i \partial_t w \, d\Omega, & q_i^{(c)} &= - \int u_i u_j \partial_j w \, d\Omega, \\ q_i^{(p)} &= - \int p \partial_i w \, d\Omega, & q_i^{(\nu)} &= \int u_i \nabla^2 w \, d\Omega. \end{aligned} \quad (16)$$

Thus neither temporal nor spatial derivatives are taken directly from the observed fields. Nine trajectories are used for discovery, three additional trajectories for validation, and three remain untouched until final testing. One hundred bootstrap refits resample complete trajectories rather than individual weak rows. SPIDER selects

$$\partial_t \mathbf{u} + 1.003459(\mathbf{u} \cdot \nabla) \mathbf{u} + 1.003470 \nabla p - 0.00501841 \nabla^2 \mathbf{u} = 0. \quad (17)$$

The DNS viscosity is 0.005 and is consulted only after discovery. The maximum coefficient error is 0.368%; validation and test residuals are 9.51×10^{-4} and 9.70×10^{-4} , respectively. The next-best support has residual 3.86×10^{-2} , and all 100 trajectory-bootstrap refits select the same four terms. SPIDER discovers the momentum equation in Eq. (17); incompressibility,

$$\nabla \cdot \mathbf{u} = 0, \quad (18)$$

is evaluated separately as a physical constraint. The pressure–Poisson and kinetic-energy relations used in later ablations are algebraically derived from the accepted momentum equation rather than independently discovered.

3.3.2. Equation-Conditioned First-Frame Representation

The downstream label is a short-horizon kinetic-energy-decay quantile, so the first-frame energy balance supplies information that a purely spatial image feature can discard. For the first observed frame, let $K = \frac{1}{2} \langle |\mathbf{u}|^2 \rangle$ and define

$$e_m(x) = \frac{\langle \mathbf{u} \cdot \mathbf{q}_m(x) \rangle}{K}, \quad s_{\mathbf{c}}(x) = c_2 e_c(x) + c_3 e_p(x) + c_4 e_\nu(x), \quad (19)$$

where $\mathbf{q}_c = (\mathbf{u} \cdot \nabla) \mathbf{u}$, $\mathbf{q}_p = \nabla p$, $\mathbf{q}_\nu = \nabla^2 \mathbf{u}$, and (c_2, c_3, c_4) are loaded from the accepted equation. By Eq. (15), $s_{\mathbf{c}}$ is the instantaneous relative kinetic-energy

decay rate. Its derivatives are computed from the first frame only; neither future frames nor class labels enter Eq. (19).

A three-layer strided 3-D convolutional encoder preserves local spatial structure before pooling. Its feature is fused with a two-layer embedding of the discovery-set-standardised scalar s_c . The spatial encoder, physics encoder, and fusion layer are jointly optimised during supervised condition pretraining and then frozen before local block training. The no-equation ablation disables the coefficient-derived scalar, the analytic ablation uses the exact Navier–Stokes coefficients, and discovered physics-informed NoProp uses the coefficients in Eq. (17).

3.3.3. Reconstructing Physical Fields from Latent Variables

Full momentum residuals require a temporal field rather than a single reconstructed snapshot. We therefore use the shared decoder

$$\mathcal{D}_\phi : \mathbb{R}^{128} \rightarrow \mathbb{R}^{9 \times 4 \times 16 \times 16 \times 16}, \quad (20)$$

which jointly reconstructs nine velocity–pressure frames. The decoder is pre-trained against trajectory windows and then frozen. Stored discovery-set means and standard deviations return its outputs to physical units before weak integration, keeping the coefficients in Eq. (17) in the units in which SPIDER measured them.

3.3.4. Weak-Form Physics Loss for Each Block

For block t , test function k , and velocity component i , let $q_{t,k,i}^{(m)}$ denote the four weak columns in Eq. (16) evaluated on the decoded time window. We define the coefficient-normalised momentum residual

$$r_{t,k,i}^{\text{NS}} = \sum_{m=1}^4 c_m q_{t,k,i}^{(m)}, \quad \eta_{t,k,i}^{\text{NS}} = \frac{r_{t,k,i}^{\text{NS}}}{\sum_m |c_m q_{t,k,i}^{(m)}| + \epsilon}. \quad (21)$$

The coefficients c_m are loaded from the validated artifact. Incompressibility is evaluated separately using $r^{\text{div}} = -\int \tilde{\mathbf{u}} \cdot \nabla w \, d\Omega$ and the analogous contribution-wise normalisation η^{div} . The main local physical loss is

$$\mathcal{L}_{\text{phys}}^{(t)} = \text{mean}_{k,i} |\eta_{t,k,i}^{\text{NS}}|^2 + 0.25 \text{mean}_{k,\tau} |\eta_{t,k,\tau}^{\text{div}}|^2. \quad (22)$$

The relation ablations add the algebraically derived pressure–Poisson and kinetic-energy residuals only in their corresponding variants. They are not treated as additional SPIDER discoveries.

3.3.5. Overall Training Objective and Algorithm

The conceptual sum of the isolated block objectives is

$$\mathcal{L}_{\text{local}} = \sum_{t=1}^T \left(\mathcal{L}_{\text{diff}}^{(t)} + \lambda \mathcal{L}_{\text{phys}}^{(t)} \right). \quad (23)$$

We implement the cosine schedule in the generative direction. The increasing signal fractions and the local training marginal satisfy

$$0 = \gamma_0 < \dots < \gamma_T = 1, \quad z_t = \sqrt{\gamma_t} u_y + \sqrt{1 - \gamma_t} \epsilon.$$

Block t predicts u_y with a uniformly weighted local squared error. At inference, deterministic coefficients are chosen as

$$b_t = \sqrt{\frac{1 - \gamma_{t+1}}{1 - \gamma_t}}, \quad a_t = \sqrt{\gamma_{t+1}} - b_t \sqrt{\gamma_t}, \quad c_t = 0. \quad (24)$$

With a perfect local prediction, this update has exactly the training marginal at γ_{t+1} ; hence the first block sees pure noise in both training and inference and the final block reaches a clean target embedding.

At each local update we sample $J \sim \text{Uniform}\{1, \dots, T\}$ and optimise $T(\mathcal{L}_{\text{diff}}^{(J)} + \lambda \mathcal{L}_{\text{phys}}^{(J)})$, an unbiased estimator of Eq. (23). Block J has its own optimiser and all other blocks receive no gradient. The spatial encoder, physical-rate encoder, fusion layer, label embedding, and decoder are pretrained once and frozen. After local block training, the classifier is trained on z_T detached from every block, so its loss cannot propagate into the local learners.

The training procedure is summarised in Algorithm 1.

3.4. Implementation Details

Choice of test functions. Following [19, 23], we use time-space tensor-product test functions:

$$w(\mathbf{x}, t) = (1 - \tilde{t}^2)^\beta \prod_{\xi \in \{x, y, z\}} (1 - \tilde{\xi}^2)^\beta. \quad (25)$$

Every local coordinate is mapped to $[-1, 1]$. Discovery and training use $\beta = 4$ with deterministic cosine modulations. The temporal derivative, spatial gradient, and Laplacian weights required by the discovered momentum relation are evaluated analytically.

Algorithm 1 Training of Physics-Informed NoProp

Require: Training data, T blocks, validated SPIDER artifact, pretrained encoder and fusion modules, label embedding, and frozen decoder $\mathcal{D}_\phi$.

- 1: **for** each local update **do**
 - 2: Sample mini-batch $\mathcal{B}$ and block $J \sim \text{Uniform}\{1, \dots, T\}$.
 - 3: Compute detached condition $E(x, s_c(x))$ and target embedding u_y .
 - 4: Sample the input latent from block J 's matching noise-to-signal marginal.
 - 5: Predict locally with θ_J and decode its output using frozen $\mathcal{D}_\phi$.
 - 6: Load the discovered coefficients and compute Eq. (22).
 - 7: Update only θ_J using $T(\mathcal{L}_{\text{diff}}^{(J)} + \lambda \mathcal{L}_{\text{phys}}^{(J)})$.
 - 8: **end for**
 - 9: Freeze all blocks and train θ_{out} on detached final latents z_T .
-

Integration domains. Discovery samples two nine-frame windows and four 16^3 spatial domains per trajectory. Eight test functions and all three momentum components yield 192 weak rows per trajectory. Training applies the same eight-function bank to each decoded nine-frame 16^3 field.

Derivative computation. Equation (16) moves every derivative in the discovered momentum relation onto w . Consequently, SPIDER discovery and the main momentum–continuity loss use field products and contractions with fixed test-function weights, without differentiating the DNS observations or decoder outputs. The optional pressure–Poisson ablation constructs a fixed test-function Hessian once on the grid, while the optional kinetic-energy ablation evaluates its dissipation term with differentiable grid gradients of the decoded velocity.

Decoder and condition architecture. A compact transposed-convolutional decoder maps $z_t \in \mathbb{R}^{128}$ to $9 \times 4 \times 16^3$ values and is frozen after pre-training. The spatial condition encoder uses three $3 \times 3 \times 3$ convolutions with channel widths 16, 32, and 64, followed by 2^3 adaptive pooling and a 128-dimensional projection. A two-layer 32-unit multilayer perceptron embeds Eq. (19), and a linear fusion layer returns the 128-dimensional condition.

Training. We use $T = 10$ independently optimised blocks, AdamW with learning rate 10^{-3} and weight decay 10^{-4} , batch size 32, mixed precision, and TF32. Each block receives 100 local updates with $\lambda = 10^{-2}$; the classifier is then trained for 30 epochs on detached final latents. Matched comparisons

use seeds $\{42, 123, 456\}$ and the same architecture, data splits, and optimisation protocol.

Extension to continuous time. NoProp also admits a continuous-time variant (NoProp-CT) [10] where the number of blocks tends to infinity and the evolution of z_t is described by a stochastic differential equation. In that setting, the physics-informed loss can be defined analogously by sampling times $t \sim \mathcal{U}(0, 1)$ and evaluating the weak residuals at those times. The diffusion loss becomes an integral over time involving $\text{SNR}'(t)$. For brevity, we focus on the discrete-time formulation; the continuous-time extension is straightforward.

4. Experiments

We conduct a comprehensive empirical evaluation of the proposed physics-informed NoProp framework on a challenging benchmark of turbulent fluid flow. Section 4.1 describes the dataset and pre-processing. Section 4.2 details the experimental setup, including baseline methods, evaluation metrics, and implementation details. Section 4.3 presents the main results on classification accuracy and physical consistency. Section 4.4 investigates robustness to noise. Section 4.5 provides ablation studies on the influence of the physics loss and the choice of discovered equations. Section 4.6 analyses the learned latent representations. Section 4.7 discusses computational efficiency and scalability.

4.1. *Dataset: Independently Generated Decaying Homogeneous Isotropic Turbulence*

We generate 15 independent trajectories of unforced decaying homogeneous isotropic turbulence by solving the incompressible Navier–Stokes equations in a triply periodic cube of side 2π . Each direct numerical simulation uses a 64^3 grid, viscosity $\nu = 0.005$, time step $\Delta t = 0.002$, initial kinetic energy 0.5, and an initial spectrum peaked at wavenumber 4. After 32 warm-up steps, 17 consecutive velocity–pressure frames are stored at every solver step. Across the trajectories, the initial Taylor-scale Reynolds number lies between 47.8 and 48.6. Automated quality gates verify monotonic kinetic-energy decay, maximum CFL below 0.062, and spectral divergence RMS below 1.5×10^{-16} ; the trajectory-level diagnostics are shown in Figure 1.

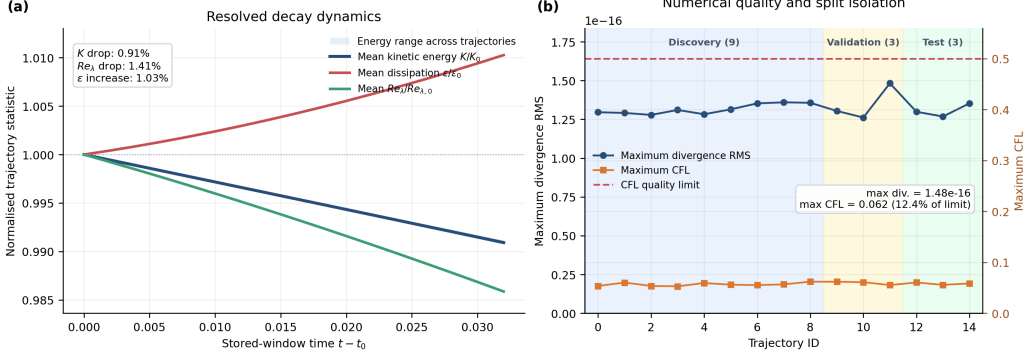


Figure 1: Quality diagnostics for the 15 independently generated decaying-HIT trajectories. (a) Mean kinetic energy, dissipation, and Taylor-scale Reynolds number across the stored window, each normalised by its first stored value; faint curves and shading show the individual energy trajectories and their range. (b) Maximum spectral divergence RMS and CFL number for every trajectory, with background bands marking the trajectory-disjoint discovery, validation, and test splits. The dashed line is the CFL quality limit used during generation.

Trajectory separation and learning samples. Complete trajectories are split before either weak domains or learning crops are constructed: nine trajectories are assigned to discovery/training, three to validation, and three to final testing. From each trajectory we draw 64 random nine-frame windows with spatial extent 16^3 , giving 960 samples. Each frame has four channels (u_x, u_y, u_z, p) . The model receives only the first frame, whereas the target is the relative local kinetic-energy decrease between the first and ninth frames,

$$d = \frac{K_0 - K_8}{\max(K_0, 10^{-12})}, \quad K_\tau = \frac{1}{2} \langle |\mathbf{u}_\tau|^2 \rangle. \quad (26)$$

Samples are divided into low- and high-entropy groups using the median initial local enstrophy of the discovery split; these groups represent different initial turbulence intensities rather than different spatial locations. Representative central cross-sections from both groups are shown in Figure 2. Within each group, discovery-split quantiles of d define five classes and are then applied unchanged to validation and test samples. Per-channel means and standard deviations are likewise computed from discovery trajectories only and reused for every split.

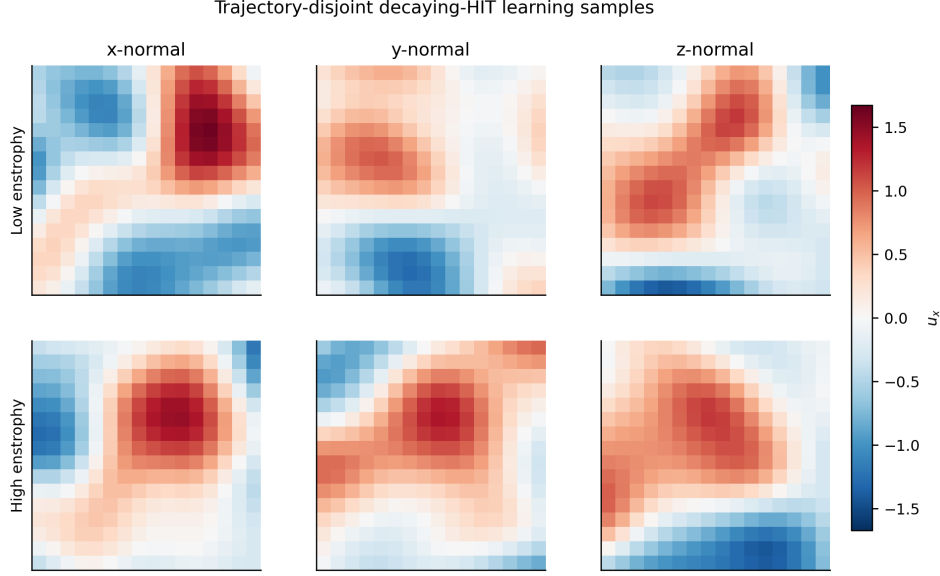


Figure 2: Representative first-frame streamwise-velocity fields from the low- and high-entropy groups. Columns show central planes normal to the x , y , and z directions using a shared colour scale. The grouping is determined by initial local enstrophy, not by spatial position.

Controlled Gaussian observation noise. Both robustness studies use additive zero-mean Gaussian noise scaled by a discovery-set field standard deviation:

$$f_{\sigma}(\mathbf{x}, t) = f(\mathbf{x}, t) + \sigma s_f \epsilon_f(\mathbf{x}, t), \quad \epsilon_f \sim \mathcal{N}(0, 1). \quad (27)$$

For clean-trained prediction, noise is added to the standardised test input at $\sigma \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$ channel standard deviations; the first-frame physical terms are recomputed from the same corrupted input, and five noise realisations are averaged for every model seed. For equation-discovery robustness, noise is instead added to the physical DNS fields before weak-feature construction at $\sigma \in \{0, 0.001, 0.005, 0.01, 0.02, 0.05, 0.1, 0.5, 1.0\}$ per-field standard deviations. The two grids resolve the different failure scales of downstream prediction and coefficient recovery and are reported separately.

4.2. Experimental Setup

4.2.1. Baseline Methods

We organise the comparison into architecture-matched physics ablations and broader learning baselines. The three matched local learners are:

- **No-equation NoProp:** the complete PI-NoProp architecture with the coefficient-derived condition disabled and $\lambda = 0$. This is the primary structure-matched control.
- **Analytic-NS NoProp:** the exact coefficients $(1, 1, 1, -0.005)$ are used in both the first-frame condition and temporal weak loss.
- **PI-NoProp (discovered NS):** the validated SPIDER coefficients are loaded from the discovery artifact into the same two paths.

We additionally evaluate five broader baselines:

- **Reference NoProp (3-D):** a one-hot NoProp adaptation of [10] in which every block owns an independent 3-D field encoder, denoiser, and optimiser. It has no supervised shared pretraining or physical input.
- **NoProp-CT (3-D conditional flow):** a continuous-time adaptation that learns the velocity of a Gaussian-to-one-hot conditional interpolation by simulation-free flow matching and uses ten-step RK4 integration at inference.
- **CNN (BP):** a supervised 3-D convolutional classifier trained on the first frame by global back-propagation.
- **Global physics-informed BP:** an encoder-classifier using the same discovered first-frame rate and temporal decoder as PI-NoProp, trained end-to-end with classification, reconstruction, and weak-physics losses. This is the implemented PINN-style global-gradient comparison [14].
- **SPIDER rate + classifier:** multinomial logistic regression fitted to the single discovered first-frame rate in Eq. (19). This deliberately strong interpretable baseline uses no future information.

All eight methods use the same trajectory-disjoint samples and seeds $\{42, 123, 456\}$, but they are not falsely presented as parameter matched. Their fitted parameter counts are approximately 3.71M for each matched PI-NoProp variant, 3.04M for reference NoProp, 0.313M for NoProp-CT, 0.137M for CNN (BP), 1.41M for global physics-informed BP, and 10 for the scalar SPIDER classifier.

4.2.2. SPIDER-Discovered Equations

SPIDER uses the five-term vector candidate set

$$\{\partial_t \mathbf{u}, (\mathbf{u} \cdot \nabla) \mathbf{u}, \nabla p, \nabla^2 \mathbf{u}, \nabla |\mathbf{u}|^2\}. \quad (28)$$

Support selection uses only the nine discovery trajectories; three validation trajectories determine acceptance and three test trajectories are reserved for final reporting. The accepted equation is

$$\partial_t \mathbf{u} + 1.003459(\mathbf{u} \cdot \nabla) \mathbf{u} + 1.003470 \nabla p - 0.00501841 \nabla^2 \mathbf{u} = 0. \quad (29)$$

Its discovery, validation, and test weak residuals are 8.72×10^{-4} , 9.51×10^{-4} , and 9.70×10^{-4} ; the maximum coefficient error is 0.368%, the next support is separated by a factor of 40.6, and all 100 trajectory-bootstrap refits retain the four physical terms. The DNS viscosity is 0.005 and is consulted only after discovery. Figure 3 compares the recovered coefficients with the analytic DNS values and shows the independent residual checks.

The main physical loss combines the discovered momentum relation with the separately evaluated incompressibility constraint $\nabla \cdot \mathbf{u} = 0$. The pressure-Poisson and kinetic-energy residuals in the relation ablation are algebraically derived from Eq. (29); they are not reported as independent SPIDER discoveries.

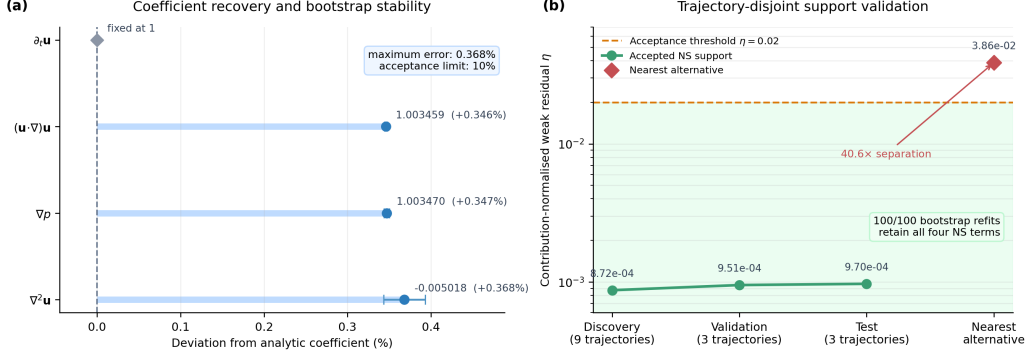


Figure 3: Recovery and independent validation of the full momentum equation. (a) Signed deviations of the SPIDER coefficients from their analytic DNS values, with one-standard-deviation trajectory-bootstrap error bars; labels give the recovered coefficients and relative deviations, while the time-derivative coefficient is fixed by normalisation. (b) Weak residuals on the disjoint discovery, validation, and test trajectory sets and for the nearest alternative support. The accepted support remains below the validation threshold, is separated from the alternative by a factor of 40.6, and is retained by all 100 trajectory-bootstrap refits.

4.2.3. Implementation Details for PI-NoProp

Network architecture. The three matched PI-NoProp variants use identical 128-dimensional condition and latent spaces and $T = 10$ independently optimised denoising blocks. A three-layer 3-D convolutional encoder processes the first 4×16^3 field, a two-layer multilayer perceptron embeds the discovered scalar rate, and a linear layer fuses both features. The shared transposed-convolutional decoder maps a block latent to $9 \times 4 \times 16^3$ velocity-pressure values. Low- and high-entropy groups use the same architecture.

Training details. All neural models use AdamW with learning rate 10^{-3} , weight decay 10^{-4} , and batch size 32. For the matched variants, the condition modules are pretrained for at most 30 epochs with validation early stopping, the temporal decoder for 40 epochs, and the label embeddings are aligned to discovery-set class centroids. These shared components are then frozen. Each NoProp block receives 100 local updates with $\eta = 0.1$ and $\lambda = 10^{-2}$ for the physical variants; the classifier is trained for 30 epochs on detached final latents. The weak loss uses $\beta = 4$, eight fixed test functions, and nine-frame 16^3 domains.

Reference NoProp likewise receives 100 updates for each of ten indepen-

dent one-hot blocks but has no shared pretraining. NoProp-CT receives 1000 conditional-flow updates and uses fixed-step ten-step RK4 inference. CNN (BP) is trained for 40 epochs; global physics-informed BP is trained for at most 30 epochs with patience eight using $\mathcal{L}_{\text{CE}} + 0.05\mathcal{L}_{\text{rec}} + 0.01\mathcal{L}_{\text{phys}}$. The SPIDER scalar baseline uses standardisation followed by multinomial logistic regression with $C = 100$.

Evaluation metrics. We report:

- **Classification accuracy:** trajectory-disjoint top-1 test accuracy, reported as mean and sample standard deviation over three training seeds.
- **Physical consistency:** for every decoded weak row, the discovered-coefficient momentum residual is normalised contribution-wise:

$$\eta_{k,i}^{\text{NS}} = \frac{\sum_{m=1}^4 c_m q_{k,i}^{(m)}}{\sum_{m=1}^4 |c_m q_{k,i}^{(m)}| + \epsilon}, \quad \eta_{\text{NS}} = \sqrt{\text{mean}_{k,i} |\eta_{k,i}^{\text{NS}}|^2}. \quad (30)$$

The reported value is batch-averaged over test samples; η_{div} is formed analogously from the three weak divergence contributions. Relation ablations additionally report η_{PP} and η_{energy} . A dash is used when a baseline does not reconstruct a temporal field.

- **Noise robustness:** clean-trained predictive accuracy and η_{NS} under the seven-level observation-noise protocol, and SPIDER support, coefficient error, bootstrap support, and held-out residual under the separate nine-level discovery-noise protocol.
- **Computational efficiency:** measured wall-clock training time, fitted parameter count, and peak allocated GPU memory. Timings are labelled by training phase rather than treated as identical epoch budgets across unlike methods.

Figure 4 reports optimisation diagnostics from the six current PI-NoProp runs without additional training or test-set tuning. Over 100 updates per block, the diffusion term decreases by nearly two orders of magnitude and the total local objective by more than one order of magnitude, while the normalised physics term remains bounded near its initial scale. After the

local blocks are frozen, both low- and high-entrophy validation accuracies stabilise near 90% within the first few classifier epochs.

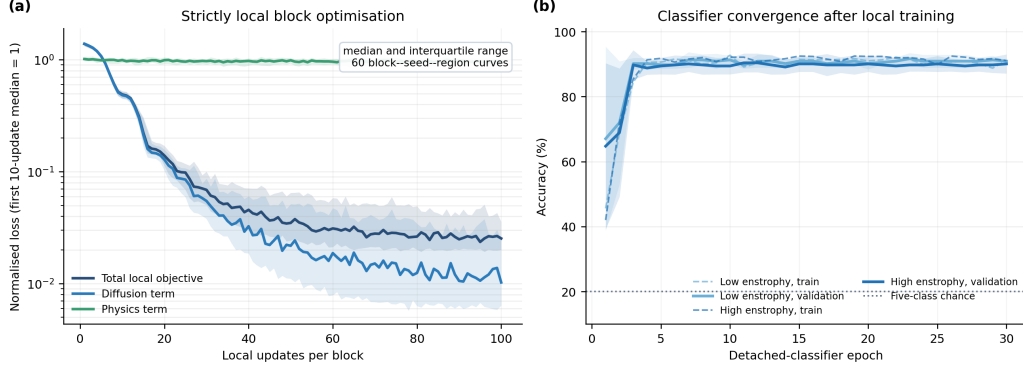


Figure 4: Training convergence of the current PI-NoProp runs. (a) Median normalised total, diffusion, and physics losses over 60 block-seed-region curves (ten blocks, three seeds, and two entrophy groups); shading denotes the interquartile range, and every curve is normalised by its own median over the first ten updates. (b) Mean training and validation accuracy of the detached classifier over three seeds, with validation shading denoting one sample standard deviation. The dotted line marks five-class chance accuracy.

4.3. Main Results: Classification Accuracy and Physical Consistency

Table 1 presents the noiseless classification accuracy and physical consistency for the low- and high-entrophy test groups. All values are the mean $\pm$ sample standard deviation over three random seeds. The five-class random-guess accuracy is 20%.

Table 1: Performance comparison on turbulent-flow classification (accuracy %) and physical consistency (η_{NS}). Bold denotes the overall best value; underlining denotes the best accuracy among methods that reconstruct a temporal field. A dash indicates that the method does not reconstruct a temporal field and therefore has no corresponding residual.

Method	Low-entropy group		High-entropy group	
	Accuracy $\uparrow$	η_{NS} $\downarrow$	Accuracy $\uparrow$	η_{NS} $\downarrow$
Reference NoProp (3-D)	18.52 ± 0.93	—	21.03 ± 4.96	—
NoProp-CT	18.21 ± 2.67	—	15.08 ± 4.81	—
CNN (BP)	14.81 ± 1.85	—	17.06 ± 5.63	—
NoProp without equations	17.59 ± 4.90	0.574 ± 0.221	17.46 ± 1.82	0.753 ± 0.125
Global physics BP	80.86 ± 5.27	0.998 ± 0.001	71.03 ± 6.87	0.991 ± 0.005
Analytic-NS NoProp	86.42 ± 1.93	0.377 ± 0.150	<u>82.94 ± 1.82</u>	0.297 ± 0.010
PI-NoProp (ours)	<u>86.73 ± 2.14</u>	0.432 ± 0.107	<u>82.94 ± 3.00</u>	0.274 ± 0.043
SPIDER rate classifier	87.96 ± 0.00	—	84.52 ± 0.00	—

Figure 5 isolates the three architecture-matched variants to show the effect of equation-derived conditioning and local physical supervision independently of the broader baseline differences in Table 1.

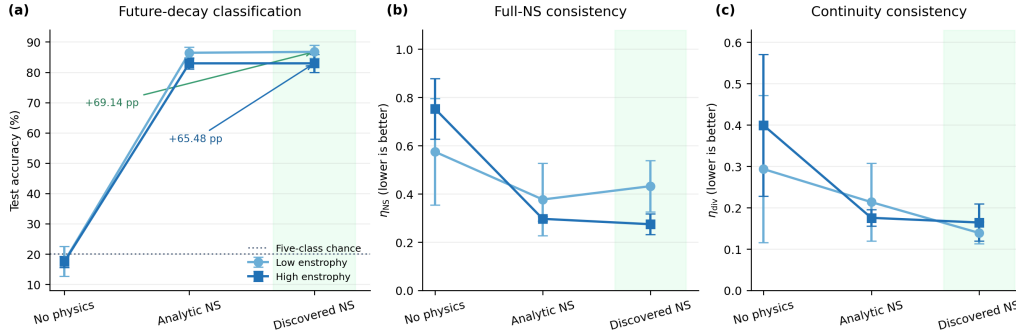


Figure 5: Architecture-matched comparison over three random seeds. (a) Five-class future-decay accuracy; the dotted line marks chance performance and annotations give the PI-NoProp improvements over NoProp without equations. (b) Contribution-normalised full-momentum residual. (c) Contribution-normalised continuity residual. Error bars denote one sample standard deviation, and the pale green bands identify the discovered-equation PI-NoProp variant.

Key observations are as follows:

- Relative to the architecture-matched NoProp without equations, PI-NoProp improves accuracy by 69.14 and 65.48 percentage points in the

low- and high-entrophy groups, respectively. Reference NoProp, NoProp-CT, CNN, and NoProp without equations remain near the 20% random-guess level, whereas Global physics BP reaches 80.86% and 71.03%. Thus, the present classification task strongly benefits from the physically derived information.

- PI-NoProp and Analytic-NS NoProp have similar predictive performance: the discovered-equation version is 0.31 percentage points higher in the low-entrophy group and tied with the analytic version in the high-entrophy group. Among the architecture-matched variants, Analytic-NS NoProp has the lowest low-entrophy η_{NS} , whereas PI-NoProp has the lowest high-entrophy η_{NS} . The results therefore support comparable utility of the discovered and analytic momentum equations rather than uniform superiority of either one.
- The ten-parameter SPIDER rate classifier obtains the highest accuracy, exceeding PI-NoProp by 1.23 and 1.59 percentage points. This result shows that the future kinetic-energy-decay label is strongly associated with the discovered physical-rate features. PI-NoProp consequently should be interpreted as integrating those discovered physics into a local-learning model, not as outperforming the specialised scalar classifier on this particular target.

4.4. Robustness to Noise

We separate corruption of a clean-trained predictor from corruption of the fields used for equation discovery. These protocols have different noise grids and failure scales and are therefore not combined into a single robustness claim.

4.4.1. Clean-Trained Prediction under Observation Noise

The PI-NoProp and architecture-matched no-equation checkpoints are trained only on clean inputs. At test time, Gaussian noise is added to the standardised first frame and the first-frame physical terms are recomputed from that same corrupted observation. Five independent observation-noise realisations are averaged for each of three model seeds, while the inference latent is fixed across noise levels and repetitions so that the varied randomness is the observation noise alone. Figure 6 reports both entrophy groups.

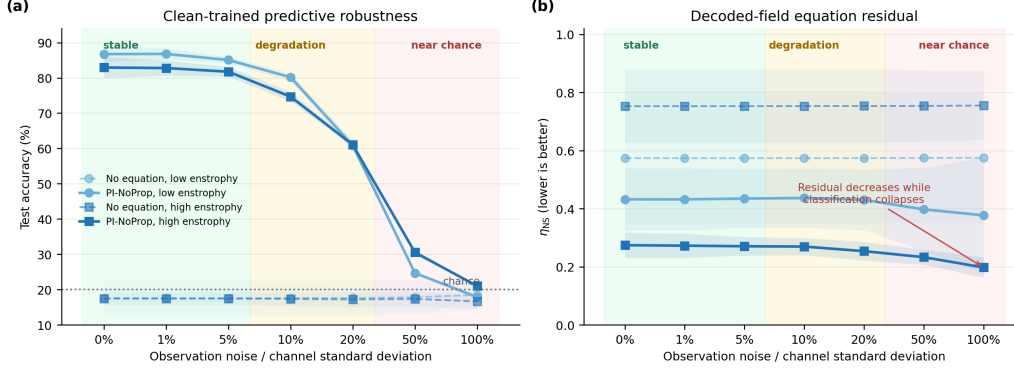


Figure 6: Clean-trained prediction under additive observation noise. (a) Five-class test accuracy; lines are three-seed means after averaging five noise realisations per seed, shading denotes one sample standard deviation across model seeds, and the dotted line marks chance performance. (b) Contribution-normalised momentum residual of the decoded fields under the same protocol. Background bands distinguish the empirically stable, degrading, and near-chance regimes; they are descriptive regions rather than additional fitted thresholds.

- At $\sigma \leq 0.05$, PI-NoProp remains close to its clean accuracy. At 5% channel-standard-deviation noise, the low- and high-enstrophy accuracies are 85.06% and 81.75%, decreases of only 1.67 and 1.19 percentage points from the clean results.
- Degradation becomes substantial beyond this range: accuracy is 80.19%/74.68% at $\sigma = 0.1$ and 60.86%/61.03% at $\sigma = 0.2$ for the low-/high-enstrophy groups. At $\sigma = 0.5$ it falls to 24.69%/30.56%, and at $\sigma = 1.0$ to 17.90%/21.03%, which is at or near five-class chance. The no-equation model is already near chance under clean input and remains so throughout the sweep.
- The decoded-field residual does not track predictive utility monotonically. For PI-NoProp, η_{NS} changes from 0.432 to 0.377 in the low-enstrophy group and from 0.274 to 0.198 in the high-enstrophy group as σ increases from 0 to 1. A smoother or more class-agnostic decoded field can have a smaller weak residual even when classification has collapsed; consequently, the high-noise residual is not interpreted as evidence of predictive robustness.

4.4.2. SPIDER Discovery under DNS-Field Noise

The second protocol adds Gaussian noise to the physical DNS velocity and pressure fields before weak-feature construction and reruns support selection and independent validation at each level. To follow the transition into failure over a wide range, this sweep uses permissive tracking gates: full four-term NS support, validation and test residuals no larger than 0.25, maximum coefficient error no larger than 200%, bootstrap support of at least 50%, and support separation of at least 1.25. These gates are used only for the corruption sweep and do not replace the stricter clean-discovery criteria in Section 4.2.2. Figure 7 reports the resulting coefficient, support, and residual diagnostics.

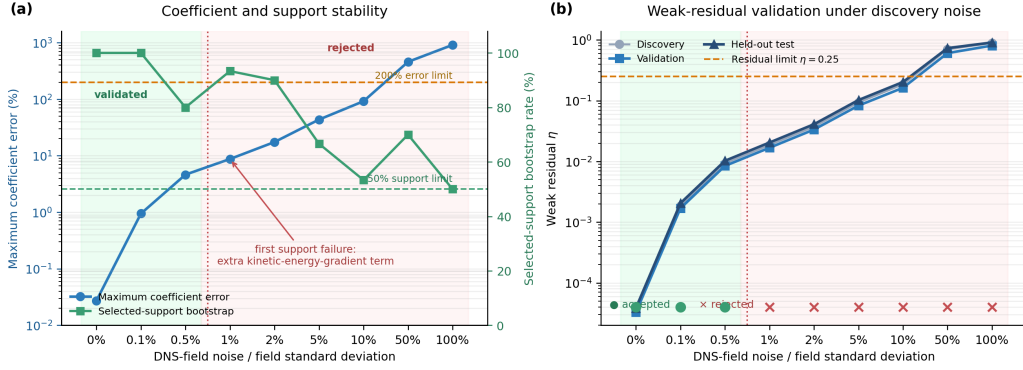


Figure 7: Robustness of SPIDER equation discovery to DNS-field noise. (a) Maximum coefficient error and trajectory-bootstrap repetition rate of the support selected at each noise level; dashed lines show the permissive corruption-sweep limits, and the vertical dotted line marks the first rejected level. Bootstrap stability alone does not imply that the repeated support is the correct four-term equation. (b) Discovery, validation, and held-out test weak residuals with the corruption-sweep residual limit. Symbols along the bottom record the overall protocol decision at each level. Noise levels are spaced categorically so that all nine tested values remain visible.

- SPIDER retains the complete momentum support at 0.1% and 0.5% field noise. The corresponding maximum coefficient errors are 0.947% and 4.63%, and the correct-support bootstrap rates are 100% and 80%, respectively. Both levels pass the corruption protocol.
- At 1% field noise, SPIDER first fails by adding the kinetic-energy-gradient candidate to the four physical terms, even though its residual and coefficient error remain below their broad tracking limits. The

same spurious term is selected at 2% and 5% noise. This identifies support selection, rather than residual magnitude alone, as the first failure mechanism.

- At 10% noise the four-term form reappears, but the support-separation ratio is only 1.04 and the result remains rejected. At 50% and 100% noise, the maximum coefficient errors rise to 458% and 913%, and the validation/test residuals also exceed the corruption-sweep limits. Thus, formal reselection of the four terms at high noise does not constitute reliable recovery of the governing equation.

4.5. Ablation Studies

4.5.1. Influence of the Physics Loss Weight λ

We vary the shared local physics-loss weight over $\lambda \in \{0, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}\}$ on the low-entropy region. The discovered momentum coefficients, data split, three seeds, and 100 updates per NoProp block are held fixed. Figure 8 reports the resulting accuracy–consistency trade-off.

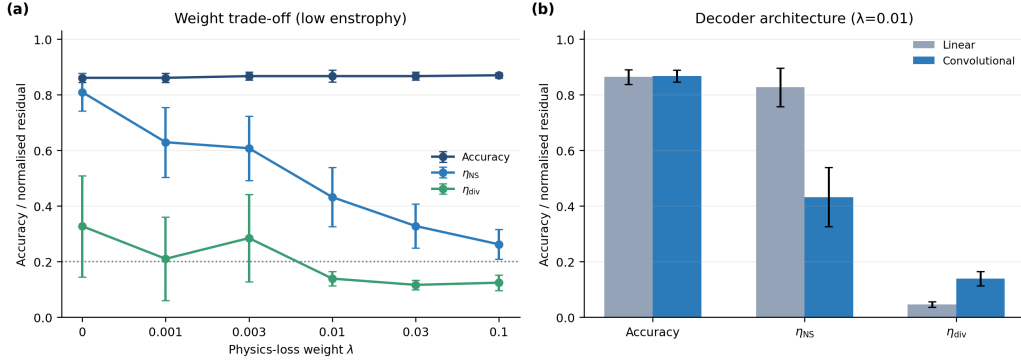


Figure 8: Ablation results for the local physics loss and temporal decoder. (a) Three-seed means and sample standard deviations for accuracy and contribution-normalised residuals across the tested λ values. Accuracy is plotted after division by 100 so that all quantities share the displayed scale; the dotted line is five-class chance. (b) Matched three-seed decoder comparison at $\lambda = 0.01$.

- The mean accuracy changes only from 86.11% at $\lambda = 0$ to 87.04% at $\lambda = 0.1$, while the NS residual decreases from 0.810 ± 0.068 to 0.262 ± 0.053 . Thus, in the tested range, increasing the physics weight improves consistency without a monotonic accuracy penalty.

- The selected value $\lambda = 0.01$ is a conservative trade-off: it gives 86.73% accuracy, $\eta_{\text{NS}} = 0.432 \pm 0.107$, and the lowest mean continuity residual among the tested values (0.139 ± 0.026). The slightly higher accuracy at $\lambda = 0.1$ is within the three-seed variability.
- This finite sweep does not support a claim about weights above 0.1 or about eventual over-regularisation; those regimes remain an explicit limitation rather than an unmeasured conclusion.

4.5.2. Choice of Physics Relations

We compare three relation sets on the low-entropy region. SPIDER independently identifies the complete momentum support; continuity is imposed as a separate incompressibility constraint, while the pressure–Poisson and energy terms are artifact-derived extensions used for ablation.

- **PI-NoProp (NS):** discovered momentum residual plus continuity.
- **PI-NoProp (NS+PP):** the NS set plus the artifact-derived pressure–Poisson relation.
- **PI-NoProp (full):** NS+PP plus the artifact-derived energy relation.

Table 2: Effect of additional physics relations on the low-entropy region ($\sigma = 0$; mean±sample standard deviation over three trajectory-disjoint seeds).

Variant	Accuracy (%)	η_{NS}	η_{div}	η_{PP}	η_{energy}
PI-NoProp (NS)	87.04 ± 1.85	0.471 ± 0.105	0.272 ± 0.092	0.777 ± 0.093	0.512 ± 0.029
PI-NoProp (NS+PP)	87.35 ± 1.93	0.375 ± 0.076	0.204 ± 0.060	0.322 ± 0.104	0.450 ± 0.008
PI-NoProp (full)	86.73 ± 1.93	0.372 ± 0.063	0.240 ± 0.046	0.251 ± 0.074	0.351 ± 0.092

Adding pressure–Poisson reduces η_{PP} from 0.777 to 0.322, and the full set further reduces it to 0.251. The energy residual similarly decreases from 0.512 to 0.351 when the energy relation is included. Accuracy remains statistically similar across the three variants, so these extensions primarily improve the corresponding decoded-field consistency rather than producing a reliable accuracy gain.

4.5.3. Effect of Decoder Architecture

We compare a direct linear latent-to-grid projection with the convolutional temporal decoder described in Section 4.2.3. The condition encoder, relation set, $\lambda = 0.01$, data split, and three seeds are matched. The linear decoder has 19.02 million parameters and obtains $86.42 \pm 2.67\%$ accuracy with $\eta_{\text{NS}} = 0.827 \pm 0.070$. The convolutional decoder has 1.23 million parameters and obtains $86.73 \pm 2.14\%$ with $\eta_{\text{NS}} = 0.432 \pm 0.107$. Its accuracy advantage is only 0.31 percentage points, but its decoded momentum residual is lower. The linear decoder has a lower continuity residual (0.046 ± 0.010 versus 0.139 ± 0.026), so the result does not justify a one-dimensional claim that convolution is uniformly superior. Because the parameter counts are not matched, this is an architecture baseline, not a capacity-controlled comparison.

4.6. Analysis of Learned Latent Representations

To analyse the learned representation without using training labels for model selection, we export the final test-set latent $z_T \in \mathbb{R}^{128}$ from the three matched variants (No equation, Analytic NS, and PI-NoProp) for all three trajectory-disjoint seeds. We fit PCA and t-SNE jointly across the variants within each enstrophy group; the projections are visualisations only, and all quantitative statistics are computed in the original 128-dimensional space. The resulting projections are shown in Figure 9.

- **Class separation:** In the low-enstrophy group, the mean pairwise class-centroid distance increases from 0.027 for No equation to 8.201 for PI-NoProp, while the 128-D silhouette score changes from -0.016 to 0.090. In the high-enstrophy group, the corresponding changes are $0.954 \rightarrow 8.575$ and $-0.043 \rightarrow 0.063$. These values support improved separation of the five decay classes.
- **Within-class spread:** PI-NoProp reduces the mean within-class covariance trace from 93.75 to 82.11 in the low-enstrophy group (12.4%), but it changes from 78.71 to 85.62 in the high-enstrophy group. Thus, the physics constraint does not uniformly make every class more compact.
- **Interpretation:** The PCA and t-SNE panels show similar grouping of the two physics-informed variants, but t-SNE geometry is projection-dependent. We therefore interpret the visualisations as evidence

of representation differences, not as proof of a smooth physical manifold or causal ordering.

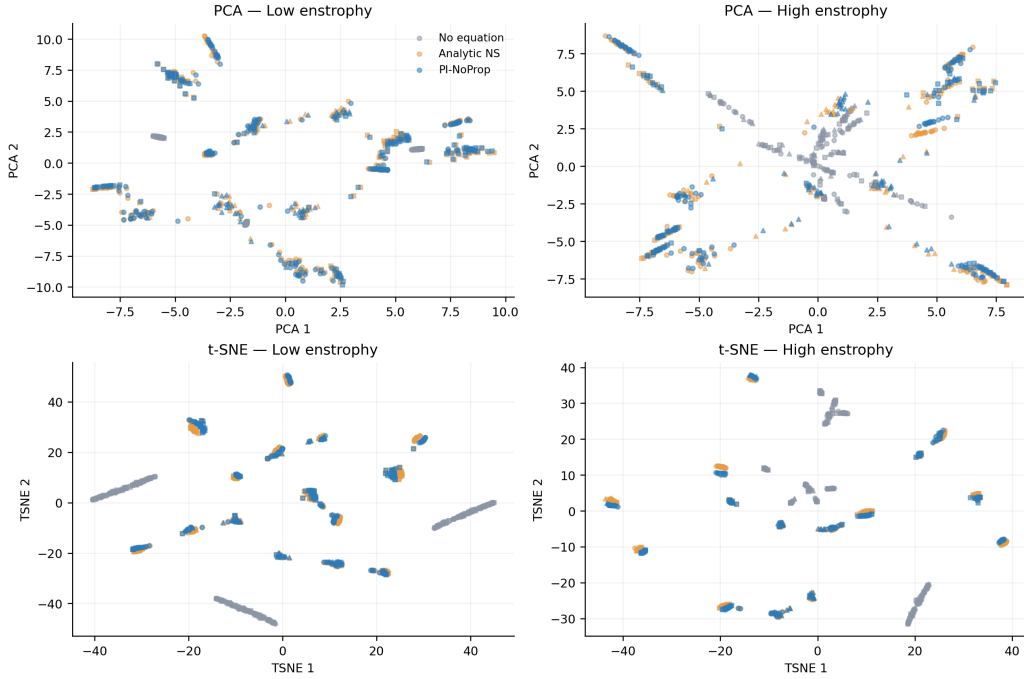


Figure 9: Held-out latent analysis using three trajectory-disjoint seeds. Columns show low- and high-enstrophy groups; rows show jointly fitted PCA and t-SNE projections of the final 128-dimensional latent. Marker shapes denote the five future-decay classes and colours denote No equation, Analytic NS, and PI-NoProp.

The measured class-separation statistics are consistent with the physics-informed condition and local loss supplying a discriminative inductive bias. The mixed within-class results also delimit the claim: the current evidence supports improved separation, not uniform compactness or a universally ordered latent manifold.

4.7. Computational Efficiency

Table 3 reports measured wall-clock training time, peak allocated GPU memory, and fitted parameter count for the low-enstrophy protocol on the NVIDIA GeForce RTX 4060 Laptop GPU used for these experiments. Each value is the mean over three seeds; training time is the recorded end-to-end

training phase for the corresponding implementation, not a claim of equal epoch budgets across methods.

Table 3: Measured computational cost on the low-entropy protocol (mean over three seeds).

Method	Training time (s)	Peak memory (MB)	Parameters
CNN (BP)	4.70 ± 0.53	89.1	137,269
Reference NoProp	10.94 ± 0.37	69.8	3,037,970
NoProp-CT	11.34 ± 0.07	28.1	313,109
No equation NoProp	12.35 ± 0.09	221.5	3,708,521
Analytic NS NoProp	33.54 ± 0.07	303.5	3,708,521
PI-NoProp	33.74 ± 0.21	303.5	3,708,521
Global physics BP	4.53 ± 1.47	308.6	1,405,161
SPIDER rate classifier	0.06 ± 0.08	< 1	10

- PI-NoProp takes longer than the no-equation control because the temporal decoder and weak-form residual are evaluated during every local update. Its measured time is close to the Analytic-NS variant, as expected from the matched architecture.
- The global physics BP entry is the implemented end-to-end weak-form baseline, not a second-order PINN measurement. It has a different optimisation schedule, so its shorter recorded training time should not be interpreted as a universal speed advantage.
- Peak memory remains below 0.31 GB for the matched PI-NoProp variants on this GPU. These numbers are hardware- and implementation-specific; they quantify the present reproducible protocol rather than establish cross-device scaling laws.

4.8. Discussion

The experimental results show that integrating the SPIDER-discovered momentum equation and its derived physical relations into NoProp’s block-wise training yields a coherent combination of predictive, physical, and computational benefits:

The central contribution is the PI-NoProp interface rather than equation discovery alone. The discovered coefficients enter the model through two coordinated paths: a learned first-frame physical condition is fused into every

NoProp block, and a frozen temporal decoder converts each block latent into a nine-frame field window whose weak momentum and continuity residuals are evaluated within that block’s local objective. The first path supplies target-free physical information before denoising, whereas the second path shapes the latent representation after each local update. Because the shared encoders and decoder are frozen during blockwise optimisation, this coupling preserves NoProp’s locality while giving the blocks a direct physical training signal.

1. **Improved accuracy:** The physics-informed condition and local residual provide a strong inductive bias, raising five-class accuracy from the near-chance no-equation control to 86.73% and 82.94% in the low- and high-entrophy groups.
2. **Physics-aware robustness:** The weak formulation avoids differentiating noisy observations, supports validated SPIDER discovery through the low-noise regime, and preserves strong clean-trained prediction through 10% channel-standard-deviation observation noise.
3. **Physically meaningful representations:** The local physics loss lowers the measured momentum residual and produces substantially larger latent class separation in both entrophy groups, as confirmed by centroid distances and silhouette scores.
4. **Local and memory-efficient optimisation:** PI-NoProp retains strictly block-local updates and uses less than 0.31 GB peak memory in the measured protocol, while the added temporal decoder and weak-form evaluations provide an explicit and interpretable computational cost.

Limitations include the dependence on the quality of SPIDER-discovered equations (which may themselves contain small errors due to discretisation) and the additional hyperparameter λ that requires tuning. Future work could explore adaptive weighting schemes or end-to-end discovery of both the equations and the network parameters.

5. Conclusion

We have presented **physics-informed NoProp**, a novel framework that integrates data-discovered partial differential equations into the blockwise diffusion-based learning paradigm of NoProp. By combining the computational efficiency and biological plausibility of local learning with the interpretability and physical fidelity of SPIDER-discovered governing equations,

our approach addresses a fundamental limitation of purely data-driven neural network training in scientific applications: the lack of explicit mechanisms to enforce consistency with the underlying physical laws that generated the observations.

The proposed method proceeds in two stages. First, SPIDER discovers and validates the sparse four-term momentum relation on the generated HIT DNS; continuity is imposed separately, and pressure–Poisson and energy relations are retained only as artifact-derived ablations. Second, the accepted momentum coefficients enter PI-NoProp through two paths: a learned first-frame physical condition is fused into every NoProp block, and a frozen temporal decoder maps each block latent to a field window whose **weak-form physics residual** is evaluated in that block’s local objective. For a given block t with latent variable z_t , the decoder $\mathcal{D}_\phi$ reconstructs the physical fields $(\tilde{\mathbf{u}}_t, \tilde{p}_t)$.

$$\mathcal{L}_{\text{phys}}^{(t)} = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \left(\|\mathcal{R}_{\text{NS}}^{(w_k)}(\tilde{\mathbf{u}}_t, \tilde{p}_t)\|^2 + \|\mathcal{R}_{\text{cont}}^{(w_k)}(\tilde{\mathbf{u}}_t)\|^2 \right), \quad (31)$$

where $\mathcal{R}_{\text{NS}}^{(w_k)}$ and $\mathcal{R}_{\text{cont}}^{(w_k)}$ denote the weak-form residuals of the Navier–Stokes and continuity equations against test functions w_k . This loss is fully differentiable and, for the momentum and continuity terms, avoids numerical differentiation of the observed DNS fields. In the implementation, the NoProp blocks are first optimised independently with local diffusion and physics objectives, after which the blocks are frozen and a classifier is trained on detached final latents. The separate objectives are:

$$\mathcal{L}_{\text{block}}^{(t)} = \mathcal{L}_{\text{diff}}^{(t)} + \lambda_t \mathcal{L}_{\text{phys}}^{(t)}, \quad t = 1, \dots, T. \quad (32)$$

$$\mathcal{L}_{\text{cls}} = -\mathbb{E}[\log \hat{p}_{\theta_{\text{out}}}(y \mid \text{stopgrad}(z_T))]. \quad (33)$$

We validated the framework on 15 independently generated decaying homogeneous-isotropic-turbulence DNS trajectories in a periodic 64^3 box at $\nu = 0.005$ (Section 4.1). Extensive experiments yielded the following key findings:

- **Improved predictive performance.** PI-NoProp reaches 86.73% and 82.94% on the low- and high-entropy groups, respectively. It exceeds

the architecture-matched no-equation control by 69.14 and 65.48 percentage points, while the specialised ten-parameter SPIDER rate classifier remains slightly higher (Section 4.3).

- **Noise behaviour.** Under clean-trained prediction, 10% channel-standard-deviation noise yields 80.19% accuracy in the low-enstrophy group and 74.68% in the high-enstrophy group; both groups approach five-class chance by 50–100%. SPIDER retains its complete support through 0.5% DNS-field noise under the stated corruption protocol; higher levels expose support and coefficient failures (Section 4.4).
- **Physical consistency.** Relative to the no-equation control, PI-NoProp lowers the low-enstrophy NS residual from 0.574 to 0.432 and the continuity residual from 0.293 to 0.139. These residual comparisons support improved consistency, but they do not establish that every physical metric is uniformly lower across all seeds or regions (Section 4.3).
- **Interpretable latent space.** PCA and t-SNE projections, together with 128-D centroid and silhouette statistics, show improved separation of the five decay classes for the physics-informed variants. Within-class spread is not uniformly reduced, so the result is interpreted as a separation effect rather than proof of a smooth ordered manifold (Section 4.6).
- **Computational efficiency.** PI-NoProp preserves block-local optimisation and uses less than 0.31 GB peak memory in the measured protocol, but its physics evaluations make it slower than the lightweight CNN and global weak-form baseline. The efficiency result is therefore a locality and memory result, not a universal wall-clock speed advantage (Section 4.7).

Ablation studies (Section 4.5) show that increasing λ over the tested range improves the NS residual without a clear accuracy penalty, while additional pressure–Poisson and energy relations mainly improve their corresponding consistency metrics. The decoder comparison gives the convolutional decoder a lower NS residual at similar accuracy, but the parameter counts are not matched and the linear decoder has lower continuity residual; therefore, the result is an informative architecture baseline rather than evidence of uniform architectural superiority.

Limitations. Several limitations of the current work should be acknowledged. First, the quality of the physics constraint depends on the accuracy

of the SPIDER-discovered momentum equation and the numerical quality of the generated DNS; the pressure–Poisson and energy relations are derived extensions whose reliability should be assessed for each dataset. Second, the hyperparameter λ_t balancing the physics and diffusion losses requires tuning; our sweep covers 0 through 0.1, while broader ranges and adaptive weighting schemes remain possible future improvements. Third, the current framework assumes that the discovered relation is valid globally across the analysed HIT domain. In systems with strong spatial heterogeneity (e.g., multiphase flows or materials with interfaces), region-specific or parametric equations may be necessary.

Future work. The synergy between local diffusion-based learning and automated equation discovery opens several promising research directions:

- **End-to-end discovery and learning.** Rather than discovering equations with SPIDER in a separate pre-processing step, one could integrate the sparse regression directly into the training loop, enabling simultaneous optimisation of the network parameters and the discovery of the governing PDEs. This would allow the equations to adapt to the representations learned by the network, potentially uncovering new physics.
- **Adaptive physics weighting.** Develop meta-learning or Bayesian approaches to automatically tune the physics loss weights λ_t based on validation performance or uncertainty estimates, reducing the need for manual cross-validation.
- **Extension to other physical systems.** Apply physics-informed No-Prop to a broader range of scientific domains, such as active matter [41, 42], plasma physics [45], climate modelling, and biomedical fluid dynamics, where first-principles models are often incomplete or unknown.
- **Handling parametric and heterogeneous systems.** Generalise the framework to systems with spatially or temporally varying coefficients by representing them as linear combinations of basis functions [43] and discovering the coefficient fields simultaneously.
- **Theoretical analysis.** Investigate the regularisation effect of the weak-form physics loss from the perspectives of statistical learning the-

ory and dynamical systems, potentially linking it to notions of stability, generalisation, and robustness.

- **Integration with continual learning.** Since NoProp naturally supports blockwise training without global gradient flow, it is well-suited for continual learning scenarios. Incorporating physical constraints could further mitigate catastrophic forgetting by anchoring the representations to invariant physical laws [44].

In conclusion, physics-informed NoProp demonstrates that **local learning and global physics can coexist and reinforce each other**. By embedding the discovered momentum equation as a soft condition and local constraint in a diffusion-based blockwise architecture, we obtain a model that combines strong predictive accuracy and physical consistency with block-local optimisation and moderate memory usage. We believe this work represents a significant step toward the broader goal of **scientifically grounded deep learning** – a paradigm where neural networks are not just predictive tools, but active participants in the discovery and representation of physical knowledge.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Complete experimental data results

Table A.1: SPIDER discovery under DNS-field noise on the generated HIT trajectories. Coefficient error is the maximum relative error over the three non-normalised coefficients; bootstrap is the selected-support repetition rate. The pass flag applies the corruption-sweep protocol, not the clean-discovery acceptance gate.

Noise	Selected support	Coeff. error (%)	Bootstrap (%)	η_{val}	η_{test}	Pass
0.0%	4-term NS	0.027	100	3.27×10^{-5}	3.85×10^{-5}	Yes
0.1%	4-term NS	0.947	100	1.69×10^{-3}	2.05×10^{-3}	Yes
0.5%	4-term NS	4.63	80	8.44×10^{-3}	1.02×10^{-2}	Yes
1%	NS + kinetic-energy gradient	8.75	93.3	1.68×10^{-2}	2.05×10^{-2}	No
2%	NS + kinetic-energy gradient	17.47	90	3.35×10^{-2}	4.10×10^{-2}	No
5%	NS + kinetic-energy gradient	43.63	66.7	8.31×10^{-2}	1.03×10^{-1}	No
10%	4-term NS (support separation 1.04)	92.04	53.3	1.63×10^{-1}	2.03×10^{-1}	No
50%	4-term NS	458.39	70	6.09×10^{-1}	7.36×10^{-1}	No
100%	4-term NS	912.87	50	8.11×10^{-1}	9.11×10^{-1}	No